LEVIATHAN SYSTEM PROMPT CHANGELOG

+ Engine 3 Quality Parity Upgrade Specification

Reference: Derailment & Failure Report (May 25-29 generation)

Date: 2026-05-25

Status: ENGINEER ACTION REQUIRED

Document Type	System Prompt Amendment + New Module Specification
Affected Systems	Engine 1 (IP Zones), Engine 3 (Trade Setups)
Change Classification	CRITICAL — Filter enforcement + module rebuild
Breaking Changes	Yes — Option expiry data flow fundamentally restructured
Backward Compat	Engine 1 output format unchanged; Engine 3 fully redesigned
Post-Mortem Grade	Engine 1: A/A+ (IP Zones) | Engine 3: D (Trade Setups)

PART A: System Prompt Changelog — Filter Enforcement & Process Hardening

These changes address the 7 derailments identified in the forensic audit. They are ordered by implementation priority. Each change includes the exact text to be added or modified in the system prompt.

CHANGE 1: Mandatory Filter Checkpoint (CRITICAL)

Addresses: Derailments #1, #2, #3, #4 | Root Cause: Over-Inclusion Bias

ADD the following as a new section in the system prompt, placed BEFORE the Protocol A execution instructions:

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MANDATORY FILTER CHECKPOINT — Execute BEFORE any option expiry data enters a report table. STEP 0: FILTER VERIFICATION GATE For every option expiry strike extracted from any data source, verify ALL of the following before the strike is written into ANY report table (Protocol A, Pillar 8, zone drivers, or day-maps). A strike that fails ANY check is EXCLUDED — no exceptions, no supplementary inclusion, no "for reference" additions. CHECK 1 — PAIR FILTER: Is the currency pair EUR/USD, GBP/USD, or AUD/USD? - YES: Proceed to Check 2 - NO: EXCLUDE. Do not enter into any table. Do not compute DXY equivalent. This includes USD/JPY, USD/CHF, USD/CAD, USD/SEK, USD/NOK, EUR/GBP, EUR/JPY, GBP/JPY, and all other crosses. Even if notional exceeds $10B. These pairs are used in K-factor DXY calculation only — they are NOT mapped as IP zone drivers. CHECK 2 — NOTIONAL FILTER: Is the notional value >= $850M (or equivalent)? - YES: Proceed to Check 3 - NO: EXCLUDE. Do not enter into any table. Do not classify as "magnet" or "wall" or "supplementary." The strike does not generate enough dealer gamma to act as a price magnet. Its DXY equivalent level is irrelevant for IP zone mapping. - EDGE CASE: If notional is $840M-$849M, EXCLUDE. The threshold is $850M. No rounding, no "approximately," no "borderline." CHECK 3 — DXY CONVERTIBILITY: Does a direct DXY equivalent exist? - YES: PASS — Strike is eligible for inclusion - NO: EXCLUDE. "N/A" or "Indirect" DXY equivalents mean the strike cannot be mapped to a DXY level and serves no IP zone mapping purpose. AFTER FILTERING: - Count total strikes that PASS all 3 checks - Count total strikes that FAIL any check - Record the Filter Audit Line: "Filter Check: R1=[pairs passed]/[total] | R2=[notional passed]/[total] | R3=[DXY convertible]/[total] | Strikes excluded: [count] | Pairs excluded: [list]" - Append Filter Audit Line to the bottom of every expiry table - Only PASS strikes proceed to classification and table entry
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Rationale: This is the single most important change. The current system prompt states the filter rules as declarative statements ("Only AU, EU, GU pairs" / "Only items >= $850M") but does not require an active verification step between data collection and data entry. The model "knows" the rules but does not "execute" them because no workflow step forces it to stop and check. This checkpoint inserts an explicit gate that must be completed before any data is written. The Filter Audit Line creates a transparency record that makes it immediately visible whether the filters were applied correctly.

CHANGE 2: Three-Tier Option Classification System (HIGH)

Addresses: Derailment #7 | Classification Corruption

REPLACE the current binary classification (CRITICAL OPTION WALL / SIGNIFICANT MAGNET) with the following three-tier system:

Tier	Label	Notional Threshold	Dealer Gamma Impact	Usage in Report
1	CRITICAL OPTION WALL	>= $1.5B	Sufficient to pin price within 10-15 pip band for 24h+	Zone driver, Option Cage, Day-map anchor
2	SIGNIFICANT MAGNET	$850M - $1.499B	Noticeable dealer hedging; may slow or deflect price but not pin	Zone driver (secondary), Day-map reference
3	EXCLUDED	< $850M	Insufficient for meaningful gamma effect	Not entered into any table

Key Changes: (a) The old "SIGNIFICANT MAGNET" tier has been split: strikes >= $850M keep the label, strikes below $850M are now "EXCLUDED" rather than relabeled. (b) A new threshold of $1.5B separates "CRITICAL OPTION WALL" from "SIGNIFICANT MAGNET" — this reflects the market reality that $1B+ notional creates meaningfully different dealer hedging behavior than $850M-$999M. (c) "EXCLUDED" is not a classification that appears in the report — it means the strike is not entered into any table at all. There is no "SUPPLEMENTARY" or "FOR REFERENCE" tier. If a strike fails the notional filter, it does not appear in the report.

CHANGE 3: Zone Synthesis Re-Verification Step (CRITICAL)

Addresses: Derailment #5 | Zone Score Contamination

ADD the following as a verification step after Zone Synthesis is complete:

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ZONE SYNTHESIS RE-VERIFICATION — Execute AFTER all zones are scored and described. For each IP Zone in the synthesis table and detail blocks: 1. Check every "Option Wall" or "Option Cage" driver reference 2. Verify that each referenced strike exists in the FILTERED expiry data (i.e., it passed Checks 1-3 in the Filter Checkpoint) 3. If any driver references a strike that was EXCLUDED by the filters: - REMOVE the driver from the zone's key drivers list - RECALCULATE the zone's pressure score without the excluded driver - RECLASSIFY the POI Tier if the score change crosses a tier boundary 4. Log any corrections: "Zone Verification: [Zone Name] — [N] excluded drivers removed, score [old] -> [new]" This step runs AFTER synthesis, not during. It is a safety net that catches any contamination that slipped through the initial filter.
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CHANGE 4: Maximum Table Density Rule (HIGH)

Addresses: Derailment #3 | Table Completeness Pressure

ADD the following rule to the system prompt formatting section:

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MAXIMUM TABLE DENSITY RULE: - Option expiry tables contain ONLY strikes that pass all 3 filter checks - If the filtered count for a day is 1 strike, the table has 1 data row - If the filtered count for a day is 0 strikes, the table shows: "No qualifying strikes (>= $850M, AU/EU/GU only) for this date" - Do NOT pad tables with sub-threshold strikes, non-qualifying pairs, or "for reference" entries to make the table appear more complete - Sparse but accurate ALWAYS beats dense but polluted - The Filter Audit Line at the bottom of each table documents how many strikes were excluded and why — this IS the completeness record
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CHANGE 5: Post-Generation Compliance Sweep (MODERATE)

Addresses: Cascade propagation | Final defense layer

ADD the following as a mandatory final step before PDF generation:

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POST-GENERATION COMPLIANCE SWEEP — Run AFTER the full report story is built. Scan every table in the report for compliance violations: 1. PAIR CHECK: No table should contain USD/JPY, USD/CHF, USD/CAD, USD/SEK, USD/NOK, EUR/GBP, EUR/JPY, GBP/JPY, or any non-AU/EU/GU pair 2. NOTIONAL CHECK: No table should contain any strike with notional < $850M (unless the table is the Missing Data Log or Filter Audit Line) 3. CLASSIFICATION CHECK: No "SIGNIFICANT MAGNET" label should appear on a strike with notional < $850M 4. ZONE DRIVER CHECK: No zone driver should reference a strike that fails R1 or R2 If any violation is found: - REMOVE the violating entry - RECALCULATE affected zone scores - LOG the correction in the Filter Audit Line - RE-RUN the sweep until zero violations remain The sweep runs until clean. No PDF is generated with violations present.
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CHANGE 6: IL -> IP Terminology Lock (ROUTINE)

Addresses: Terminology consistency

CONFIRM the following terminology is enforced throughout the system prompt and all generated output:

Old Term	New Term	Scope	Status
Institutional Liquidity (IL)	Institutional Pressure (IP)	All text, labels, table headers, zone names	MANDATORY — no exceptions
IL Zone	IP Zone	All references	MANDATORY
IL Levels	IP Levels	All references	MANDATORY
Liquidity Sweep Zone (LSZ)	LSZ (keep abbreviation)	LSZ abbreviation is acceptable; full name remains Liquidity Sweep Zone	ACCEPTABLE as-is

PART B: Engine 3 Quality Parity Upgrade Specification

Engine 3 (Trade Setup Architecture) was disabled for the May 25-29 report because its prior output quality was graded D in the post-mortem. This section diagnoses why Engine 3 failed and specifies what changes are needed to bring it to the same A/A+

quality standard as Engine 1 (IP Zones) and Engine 2 (Macro Shape Chain).

B.1: Why Engine 3 Failed — Root Diagnosis

The post-mortem of the April 20-24 report graded Engine 1 at A/A+ and Engine 3 at D. The quality gap was not marginal — it was catastrophic. The following analysis identifies the specific failure modes:

Failure Mode	Description	Impact on Grade
Garbage In, Garbage Out	Engine 3 consumed unfiltered option data (same as Engine 1 before this changelog). Sub-threshold strikes were used as entry/exit anchors, producing setups anchored to levels with no institutional gravity.	FATAL — Trade entries at levels with no magnet effect = random entries
No Confluence Floor	Engine 3 would generate a trade setup if ANY single pillar suggested an entry, regardless of whether other pillars confirmed. A setup with 2/14 pillars firing was treated the same as one with 10/14.	CRITICAL — Low-confluence setups are noise, not signals
Vague SL/TP Placement	Stop losses were placed at "nearby support" without anchoring to specific LSZ edges. Take profits were set at arbitrary round numbers instead of opposing IP zone boundaries.	CRITICAL — SL at wrong level = stopped out for no reason; TP at wrong level = leaving money on the table
No RRR Gate	Risk/reward ratios were calculated but not gated. Setups with 0.8:1 RRR were presented alongside setups with 3.5:1 RRR without any quality filter.	HIGH — Low-RRR setups waste capital and attention
No Temporal Alignment	Engine 3 did not align trade setups with the Macro Shape Chain from Engine 2. A bullish setup could be generated on a day when Engine 2 predicted a whiplash/rejection pattern.	HIGH — Right level, wrong time = stopped out
No Red Folder Deferral	Engine 3 did not incorporate the Red Folder time-locks from Protocol F. Setups were generated for times when high-impact data releases would make technical levels irrelevant.	MODERATE — Trading into NFP/PCE is gambling, not analysis

B.2: What Engine 3 Needs — Quality Parity Requirements

To match the A/A+ quality of Engine 1 and Engine 2, Engine 3 must meet the following requirements. Each requirement is derived from the failure mode it addresses and the quality standard set by the other two engines.

#	Requirement	Spec	Addresses	Priority
R3.1	Filtered Data Only	Engine 3 must ONLY use option data that passes the Filter Checkpoint (Checks 1-3). No sub-threshold strikes, no non-AU/EU/GU pairs. The same filtered dataset that feeds Engine 1 feeds Engine 3.	Garbage In, Garbage Out	CRITICAL
R3.2	Minimum Confluence Floor	A trade setup requires MINIMUM 6/14 pillars firing to be generated. Below 6/14 = no setup. The confluence floor is non-negotiable and applies to all setup types (long, short, scalp, swing).	No Confluence Floor	CRITICAL
R3.3	LSZ-Anchored Stop Losses	Every SL must be anchored to a specific LSZ edge (e.g., "SL: 98.49 (LSZ demand-side, Mar 10 Low)"). No SL at "nearby support" or "below the zone." The SL must reference the LSZ range, source date, and type.	Vague SL/TP	CRITICAL

#	Requirement	Spec	Addresses	Priority
R3.4	IP Zone-Anchored Take Profits	Every TP must be anchored to an opposing IP Zone boundary (e.g., "TP1: 99.45 (Zone #1 Resistance Ceiling bottom edge), TP2: 99.52 (Zone #1 top edge)"). No TP at round numbers or arbitrary levels.	Vague SL/TP	CRITICAL
R3.5	RRR Gate	Minimum RRR for any generated setup: 2.0:1. Setups with RRR below 2.0:1 are excluded from the report. RRR is calculated as (TP1 - Entry) / (Entry - SL). For counter-trend setups, minimum RRR: 2.5:1.	No RRR Gate	HIGH
R3.6	Macro Shape Alignment	Every Engine 3 setup must specify which Macro Shape from Engine 2 it aligns with (e.g., "Shape Alignment: SS -> WL at ceiling, then rejection -> downside entry"). If Engine 2 predicts SS and Engine 3 generates a directional breakout setup, the setup must note the DISAGREEMENT and downgrade confidence.	No Temporal Alignment	HIGH
R3.7	Red Folder Deferral	No setup entry within 30 minutes of a Red Folder event. Setups that would trigger during a Red Folder window are marked "DEFERRED — Red Folder at [time]. Resume after [time+30min]." The deferral does not invalidate the setup; it delays entry.	No Red Folder Deferral	HIGH
R3.8	Entry Precision	Entry price must be a specific DXY level, not a range or zone. (e.g., "Entry: 99.18" not "Entry: Zone #2 area"). The entry must correspond to a specific option strike DXY equivalent, LSZ edge, or MA confluence level.	Vague entries	HIGH
R3.9	Confidence Scoring	Each setup gets a confidence score based on: Confluence Pillar Count (weight 40%), Option Structure Strength (20%), Macro Shape Alignment (20%), Red Folder Clearance (10%), ZFP of anchor zone (10%). Score range: 0-100. Only setups scoring >= 65 are included.	No quality gate	MODERATE
R3.10	Setup Limit	Maximum 3 setups per report. Quality over quantity. If only 1 setup meets all criteria, generate 1. If zero setups meet all criteria, generate zero and note "No qualifying setups this week — all zones require confirmation."	Over-generation	MODERATE

B.3: Engine 3 Architecture — Proposed Workflow

The following workflow enforces all 10 requirements. It must be added to the system prompt as the Engine 3 execution specification:

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ENGINE 3: TRADE SETUP ARCHITECTURE — EXECUTION WORKFLOW Phase 1: ZONE SELECTION (Depends on Engine 1 output) - Input: IP Zone Map from Engine 1 (filtered data only) - For each zone with Pressure >= MODERATE (7/14+): - Identify whether zone is a potential ENTRY zone or TP target - Entry zones: zones where price is approaching (within 15 pips) - TP zones: opposing zones in the direction of the trade - Output: Candidate zone list with entry/TP classification Phase 2: SHAPE ALIGNMENT (Depends on Engine 2 output) - Input: Macro Shape Chain from Engine 2 - For each candidate zone from Phase 1: - Check if the predicted Macro Shape supports a trade at this zone - If shape = BS (directional breakout): directional trade supported - If shape = WL (whiplash): counter-trend fade supported, directional trade DOWNGRADED - If shape = SS (swing scalp): no directional trade; scalp only with tight RRR - If
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shape = R (reversal): counter-trend entry supported - DISAGREEMENT LOG: If Engine 2 shape contradicts the trade direction, note the disagreement and reduce confidence score by 20 points - Output: Shape-aligned candidate list Phase 3: CONFLUENCE SCORING - For each shape-aligned candidate: - Count pillars firing from the 9-Pillar system at the entry level - Minimum 6/14 pillars required to proceed - Score breakdown: * Pillar 1 (Pivots): Does entry align with a pivot level? (0 or 1) * Pillar 2 (MAs): Does entry align with an MA confluence? (0 or 1) * Pillar 3 (Structure): Does entry align with a structural level? (0 or 1) * Pillar 4 (Analyst): Does any analyst cite this level? (0 or 1) * Pillar 5 (Extremes): Is entry near a swing high/low or repeated magnet? (0 or 1) * Pillar 6 (Round Numbers): Is entry at a big/quarter figure? (0 or 1) * Pillar 7 (DXY Options OI): Does ICE DXY OI data support? (0 or 1, or 0 if no data) * Pillar 8 (FX Options): Does a filtered option strike map here? (0 or 1) * Pillar 9 (4H Zones): Does a 4H BoM/MoM zone overlap? (0 or 1) - Cross-pillar bonuses: * Option + Structure overlap: +1 * 3+ pillars in same 20-pip band: +1 * Repeated Magnet at entry: +1 - Maximum possible score: 14 - If score < 6: REJECT — no setup generated - Output: Scored candidate list with pillar breakdown Phase 4: SL/TP ANCHORING - For each scored candidate that passed Phase 3: - SL: Anchor to nearest LSZ edge on the wrong side of the trade * For longs: SL = demand-side LSZ bottom edge - 3 pips buffer * For shorts: SL = supply-side LSZ top edge + 3 pips buffer * If no LSZ exists within 30 pips: REJECT — no valid SL anchor - TP1: Anchor to nearest opposing IP Zone boundary * For longs: TP1 = next resistance zone bottom edge * For shorts: TP1 = next support zone top edge - TP2: Anchor to second opposing IP Zone boundary (if within 80 pips) - Calculate RRR: (TP1 - Entry) / (Entry - SL) * If RRR < 2.0: REJECT — insufficient reward * If counter-trend and RRR < 2.5: REJECT - Output: Anchored setups with precise SL/TP and RRR Phase 5: CONFIDENCE SCORING & RED FOLDER CHECK - For each anchored setup: - Base confidence = (Confluence Score / 14) * 40 [max 40 points] - Option bonus = (Option notional / $3.5B) * 20 [max 20 points] - Shape alignment = 20 if aligned, 0 if disagreeing [max 20 points] - Red folder clearance = 10 if no Red Folder within 2h of entry, 0 if conflict - ZFP bonus = (1 - ZFP of anchor zone) * 10 [max 10 points, lower ZFP = stronger zone] - Total: 0-100 scale - If score < 65: REJECT - If Red Folder within 30 min of entry: DEFER, do not reject - Sort by confidence score descending - Take top 3 maximum - If zero setups score >= 65: Output "No qualifying setups" message Phase 6: OUTPUT FORMATTING - Each setup rendered as a KV table with the following fields: * Direction (Long/Short) * Entry: [exact DXY level] — [source: option strike / LSZ edge / MA confluence] * Stop Loss: [exact DXY level] — [LSZ anchor: range, source date, type] * TP1: [exact DXY level] — [IP Zone anchor: zone name, boundary] * TP2: [exact DXY level] — [IP Zone anchor: zone name, boundary] (if exists) * RRR: [value]:1 * Confidence: [score]/100 * Confluence: [count]/14 pillars — [list which pillars fire] * Shape Alignment: [Engine 2 shape at entry zone] * Red Folder Status: [CLEAR / DEFERRED until time] * Anchor Zone: [Engine 1 zone name and pressure score] * Risk Note: [1-2 sentence assessment of what could go wrong]
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B.4: Engine 3 Data Dependencies

Engine 3 requires the following data inputs to produce quality-parity output. Each dependency maps to a specific requirement:

Data Input	Source	Requirement Served	Current Status
Filtered Option Expiry Data	Protocol A (post-Checkpoint)	R3.1, R3.4, R3.8	AVAILABLE after Change 1 is implemented
IP Zone Map + LSZ Edges	Engine 1 output	R3.3, R3.4, R3.8	AVAILABLE — Engine 1 produces this
Macro Shape Chain	Engine 2 output	R3.6	AVAILABLE — Engine 2 produces this
Red Folder Schedule	Protocol F	R3.7	AVAILABLE — already in report
9-Pillar Confluence Scores	Engine 1 internal state	R3.2, R3.9	NEEDS EXPOSURE — currently internal to Engine 1
CME FX Open Interest	Protocol C1/C2/C3	Pillar 7 scoring	NO DATA — CME sources blocked from environment
ICE DXY Options OI	Pillar 7 supplementary	Pillar 7 scoring	NO DATA — feed inaccessible

Critical Dependency Gap: Pillar 7 (CME FX OI + ICE DXY Options OI) consistently returns NO DATA because the data sources are blocked from this environment. This means Engine 3 can score a maximum of 12/14 confluence pillars instead of 14/14. The practical impact is that the confluence floor should be adjusted to 5/12 (proportional to the original 6/14) when Pillar 7 data is unavailable, and a note should be appended to each setup: "Pillar 7 data unavailable — confluence score capped at 12/14." If Pillar 7 data becomes available in the future, the floor returns to 6/14.

B.5: Implementation Priority & Rollout Plan

Phase	Changes	Dependencies	Target Quality	Timeline
Phase 1: Filter Fix	Changes 1-6 (Part A)	None — can be implemented immediately	Engine 1: A+ (preserved) | Engine 2: A (preserved)	Immediate

Phase	Changes	Dependencies	Target Quality	Timeline
Phase 2: Engine 3 Rebuild	Requirements R3.1-R3.10 (Part B)	Phase 1 must be complete; Engine 1+2 output format stable	Engine 3: B+ minimum, A target	After Phase 1 verified
Phase 3: Pillar 7 Data	External data feed integration	Infrastructure access to CME/ICE	Engine 1: A+ confirmed | Engine 3: A target	When infrastructure allows

Phase 1 is the only blocking change. It must be implemented and verified before Engine 3 is re-enabled. The May 25-29 report demonstrated that without the Filter Checkpoint, both Engine 1 and Engine 3 produce contaminated output. Engine 1 survived because its other pillars (structure, MAs, pivots) are strong enough to produce A-grade zones even with polluted option data. Engine 3 does not have this resilience — trade setups are almost entirely dependent on the quality of the option data anchoring, because the entry, SL, and TP levels are directly derived from it. When the option data is polluted, Engine 3 entries are anchored to levels with no institutional gravity, producing setups that look precise but are fundamentally random.

The recommended rollout: (1) Implement Part A changes immediately. (2) Re-run the May 25-29 report with the corrected filters and verify that the corrected output matches the "Corrected Data" section of the Derailment Report. (3) Only after verification is complete, implement Part B (Engine 3 rebuild). (4) Run a forward test with Engine 3 re-enabled on the next trading week. (5) Post-mortem the Engine 3 output before making it the production standard.